

NEPHELOMETRY & TURBIDIMETRY

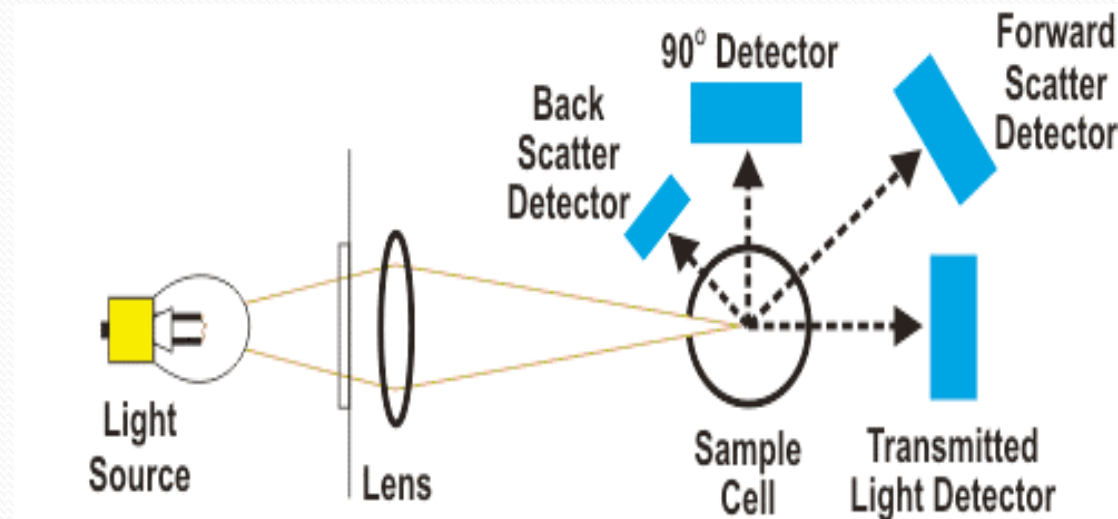
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INTRODUCTION

- When EMR(light) strikes a particle in solution, some of the light will be:
 - **Absorbed**
 - **Transmitted** through the solution
 - **Scattered or reflected** by the particle .
- The amount of light scattered is proportional to the concentration of insoluble particle.
- concept of **light scatter**

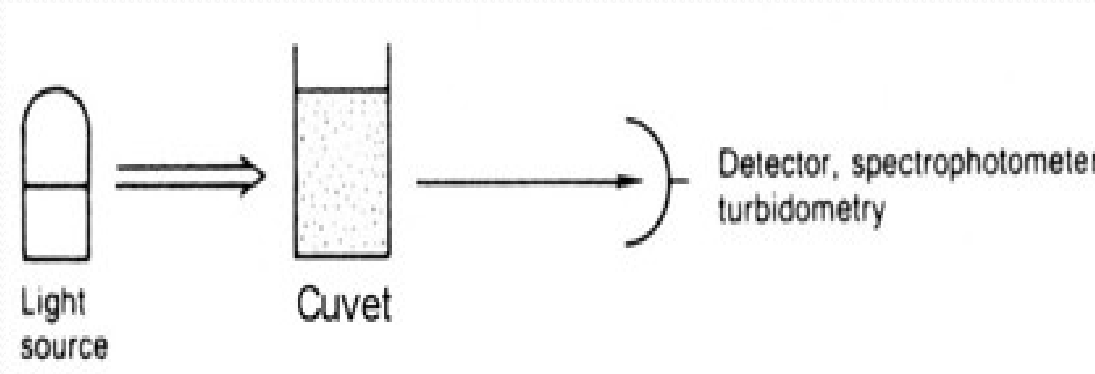
THEORY

- Scattered light may be measured by
 - Turbidimetry
 - Nephelometry
- In turbidimetry , the intensity of light transmitted through the medium, **the unscattered light**, is measured.



Cont...

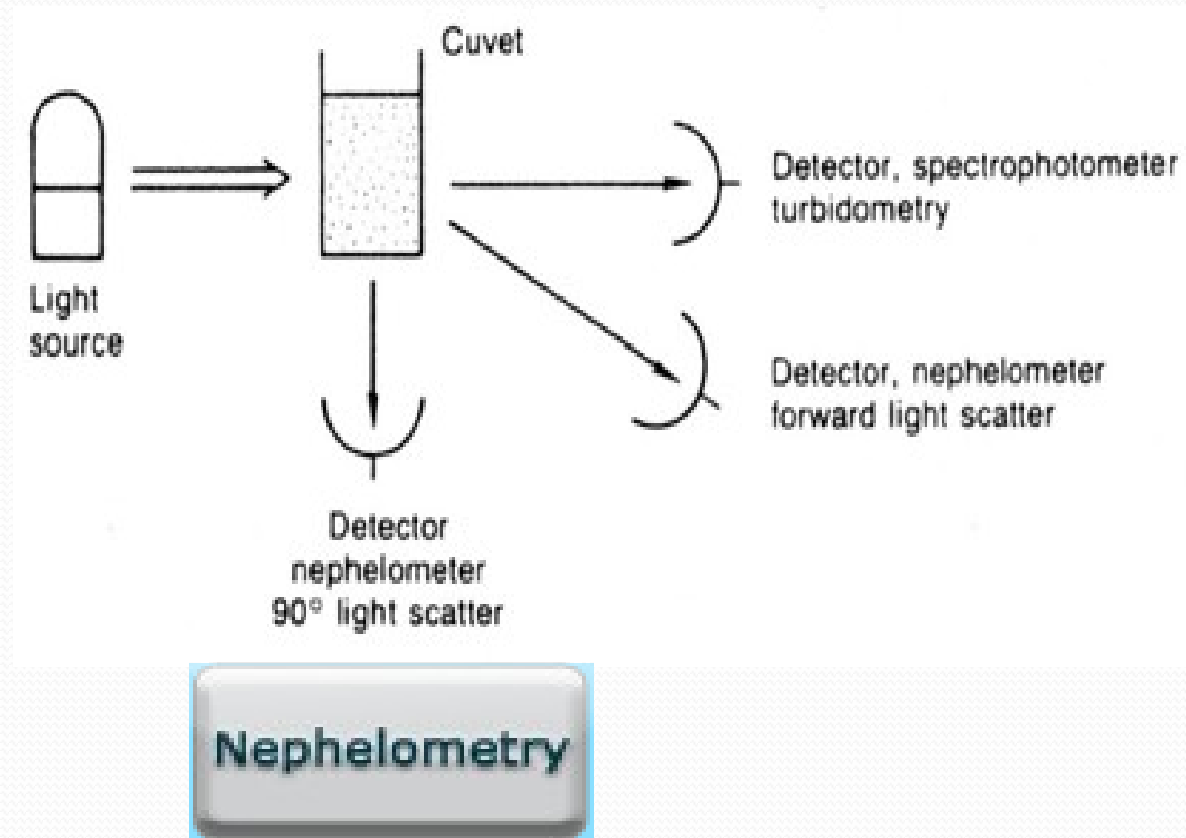
- Turbidometric measurements are made at 180° from the incident light beam.



- In Nephelometry, the intensity of the scattered light is measured, usually, but not necessarily, at right angles to the incident light beam.

Con't...

- The two techniques differ in the manner of measuring the scattered radiation.



Con't...

- Turbidity can be measured on most routine analysers by a spectrophotometer (**absorbed light**)
 - Reduced sensitivity and precision.
- The intensity of **scattered light** is normally measured by Nephelometer.

Cont...

- Light scattering is the physical phenomenon resulting from the interaction of light with a particles in solution.

Dependent on :

- Particle size
- Wavelength
- Distance of observation,
- Concentration of particles
- Mwt of particles

Con't...

- Scattering of light- by particles in a colloid or suspension.
- The longer-wavelength light is more transmitted while the shorter-wavelength light is more reflected via scattering.

Light Scattering Phenomenon

The difference in how longer and shorter wavelength light interacts with materials is primarily due to scattering mechanisms and energy levels:

1. Rayleigh Scattering (Shorter Wavelengths Scatter More)

- When light encounters particles much smaller than its wavelength, scattering intensity is proportional to $1/\lambda^4$.
- Since shorter wavelengths (blue and violet light) have higher energy, they interact more with small particles, causing them to scatter more.
 - This is why the sky appears blue—tiny air molecules scatter short-wavelength light more than long-wavelength light.

Con't...

2. Mie Scattering (Less Wavelength Dependence)

- When particles are similar in size to the wavelength of light, scattering becomes less dependent on wavelength.
- This explains why clouds appear white—water droplets scatter all wavelengths more evenly.

Con't...

3. Transmission of Longer Wavelengths

- Longer wavelengths (red and infrared light) have lower energy, making them less likely to interact with small particles.
- Instead of scattering, they tend to pass through materials more easily, which is why infrared light penetrates fog and smoke better than blue light.

Con't...

4. Material Absorption and Reflection

- Some materials preferentially absorb or reflect certain wavelengths based on their atomic structure.
- For example, glass transmits infrared light more efficiently than ultraviolet light, affecting how different wavelengths travel through materials.

Con't...

Size Parameter ($x = 2\pi r/\lambda$)

- The ratio of particle size (r) to wavelength (λ) determines the type of scattering.
- ✓ If $x < 0.1$, Rayleigh scattering dominates.
- ✓ If $x \approx 1$, Mie scattering occurs.
- ✓ If $x > 10$, geometric optics effects take over.(light behaves more like reflection and refraction)

Con't...

NEPHELOMETRY

- ↓ concentration, uniform scattering, intensity of scattered light proportional to conc. measured at 90°

TURBIDIMETRY

- ↑ concentration, scattering not uniform, intensity of transmitted light measured at 180°

Colorimeter

Turbidimeter

Similarity:

Transmitted light measured at 180°

Difference:

- Absorption of radiation
- Scattering of radiation

Flourimetry

Similarity: Nephelometry

- Emergent radiation measured at 90°

Difference:

- Emitted radiation measured
- Emitted radiation – longer WL than incident light
- Scattered radiation measured
- Scattered radiation – same WL as that of incident light

CHOICE OF THE METHOD

- ❖ Depends upon the amount of light scattered by suspended particles present in solution.
- **Turbidimetry** - high concentrated suspensions
- **Nephelometry** - low concentrated suspensions
 - more accurate results

Con't...

Turbidimetry is best suited for highly concentrated suspensions because it measures the loss of transmitted light intensity due to scattering by particles.

- ✓ 1. **Higher Particle Density**– In highly concentrated suspensions, more particles interact with light, leading to significant scattering and absorption. This makes turbidimetric measurements more reliable.
- ✓ **Direct Measurement of Transmission** – Unlike nephelometry which measures scattered light at an angle, turbidimetry detects the reduction in transmitted light at 180° , making it ideal for dense samples where scattering is extensive.

INSTRUMENTATION

The basic instrument contains

- Light Source: Tungsten lamp
- Filters
- Sample cells
- Detectors (photometric)

FACTORS AFFECTING MEASUREMENTS

The amount of radiation removed or deviated from the primary radiation beam depends on the following factors

- **Turbidimetry:**

$$S = \log I/I_0 = kbc$$

- $T = \text{Transmittance} = I/I_0$
- $S = \text{turbidity due to scattering}$
- $k = \text{turbidity constant}$
- $b = \text{path length}$
- $c = \text{concentration of suspended material}$

- **Nephelometry:**

$$I_s = K_s I_0 C$$

- $I_s = \text{scattered intensity}$
- $K_s = \text{empirical constant}$
- $I_0 = \text{Incident intensity}$
- $c = \text{concentration of suspended material}$

Con't.....

- Turbidimetry-Practical Considerations
- Selecting λ : Important.

Sample Preparation

- Scattering is related to:
 1. Conc of the scattering particles
 2. Particle size
 3. Particle shape

Applications

- A number of pharmaceutical substances are assayed either:
 - Turbidimetrically or
 - Nephelometrically

Applications...

- A large number of antibiotics,
 - chlortetracycline, doxycycline, gentamicin, neomycin, streptomycin, tobramycin and others
 - assayed turbidimetrically with fairly good accuracy
- Nephelometric assay
 - Sulphate &
 - phosphate ions efficiently

Applications...

- Selected Precipitates used in Turbidimetric & Nephelometric Methods

Analyte	Precipitant	Precipitate
Ag^+	NaCl	AgCl
Ca^{2+}	$\text{Na}_2\text{C}_2\text{O}_4$	CaC_2O_4
Cl^-	AgNO_3	AgCl
CN^-	AgNO_3	AgCN
CO_3^{2-}	BaCl_2	BaCO_3
F^-	CaCl_2	CaF_2
SO_4^{2-}	BaCl_2	BaSO_4